

UNDERSTANDING TUPLE

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TUPLE

A tuple in Python is an immutable, ordered collection of items. Tuples are similar to lists, but once created, their elements cannot be changed, added, or removed.

They are useful for storing fixed collections of items, such as coordinates, RGB color values, or constant data.

Syntax:

```
my_tuple = ( item1, item2, item3 )  
print(my_tuple)
```

TUPLE

Key Features of Tuples:

1. Immutable: Once created, the elements of a tuple cannot be modified.
2. Ordered: The elements are stored in a specific order.
3. Allow Duplicates: Tuples can have duplicate elements.
4. Heterogeneous: They can store elements of different data types.

TUPLE

1 With Parenthesis

```
tup1= (1, 2, 3, 4)
```

2. Without Parenthesis

```
tup2= 2, 3, 4, 5
```

3. Single Element Tuple

```
Single_element = (5,) # This is a tuple not_a_tuple = (5)
```

TUPLE

4. Using tuple() Constructor

```
my_tuple = tuple([1, 2, 3]) # Converting a list to a tuple  
print(my_tuple)
```


TUPLE OPERATION

a. Concatenation

```
tuple1 = (1, 2)
tuple2 = (3, 4)
result = tuple1 + tuple2
print(result)
```

b. Repitition

```
my_tuple = (1, 2)
print(my_tuple * 3) # Output: (1, 2, 1, 2, 1, 2)
```

THANK YOU

